

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects need not add. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with a synergy order parameter $\mathcal{S}$ (meV/atom). Under a rigid-strain upper-bound protocol (PBE+D3), tetramers and fifteen periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, B/N/P @ +3%, seed 42 placement) give opposite strain coefficients for N (−303 meV/%) and B (+57 meV/%) and $|\mathcal{S}|$ up to 31.9 meV/atom (P, $n=1$) at +3% strain, largest for P despite small $|\alpha|$. The pattern warns that sequential high-throughput screens can mis-estimate *relative* stability margins when local dopant distortion couples nonlinearly to external strain; Marcus λ and hopping diagnostics (Supplemental Material Fig. S5–S6) are orthogonal to the $\mathcal{S}$ audit, reported only in Supplemental Material, and remain incomplete until vertical single-points at the neutral geometry finish.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1–3]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a same-functional total-energy cross term under joint biaxial (ϵ, δ) load. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [8], so external biaxial load couples to an already distorted cage baseline. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 9–11], while related 2D fullerene networks show anisotropic mechanical and tribological response under load [12, 13] and endohedral qHPC₆₀ cage filling reshapes band gaps [14]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than additive compositional estimates. Non-additive coupling is documented in other two-dimensional networks [1]; a quantitative *total-energy* cross term $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a single PBE+D3 protocol on fifteen periodic (n, δ) points at +3% strain (seed 42 placement).

We address three questions on this model network: (1) how opposite linear strain coefficients α emerge for donor vs. acceptor dopants; (2) how large the omitted cross term $|\mathcal{S}|$ is when strain and substitution are applied jointly; and (3) whether $|\mathcal{S}|$ tracks $|\alpha|$ or instead reflects local structural distortion. Using $\gtrsim 90$ converged CP2K

tasks [15] (Table S2), we define $\mathcal{S}$ on shared tetramer and periodic grids; Marcus λ and hopping observables (IPR/ J) are orthogonal to the $\mathcal{S}$ audit, reported only in Supplemental Material Fig. S5–S6 (vertical single-points in progress; 6/8 converged at revision), and are not required for the non-additivity conclusions here. The central narrative anchor is that *local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or from strain-only elastic increments need not survive a coupled load (without reordering widely separated E_{sub}).

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All calculations use CP2K [16] with PBE [17] and Grimme DFT-D3 (Becke–Johnson damping) [18] (Supplemental Material for basis, SCF, and task inventory). We deliberately remain below the Koopmans/rVV10 level of Ref. 4 to obtain a uniform, size-converged (ϵ, δ) scan rather than absolute polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is always $\mathcal{S}$ (calligraphic) in meV/atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

Rigid-strain protocol. Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are *upper bounds* on coupling under load until the ionic-relaxation benchmark in Table S3 converges; the rigid tetramer reference cross term in Table S3 ($\mathcal{S} = +0.96$ meV/atom)

already differs in sign from the periodic $n=4$ P value (-23.7 meV/atom), underscoring that relaxation and concentration matching remain open checks.

Plane-wave cutoff (size scaling). Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table S2); comparisons at fixed n always share the same cutoff. Cross- n trends therefore mix a modest basis change at $n = 6$; we do *not* interpret the reported N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ as a physical size effect until the matched 400 Ry $n=6$ control in Supplemental Table S2 completes.

C. Substitutional doping and grids

Tetramer models ($4 \times C_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates, random seed 42), not a thermodynamic minimum. A second seed 137 tetramer ensemble (18 ENERGY tasks; Table S4) tests whether α and tetramer-scale $\mathcal{S}$ are sensitive to dopant placement; periodic $\mathcal{S}(n)$ claims remain on the seed 42 periodic protocol until seed 137 periodic points are added. Periodic scaling cells place one dopant per C_{60} ($n = 1-8$), so nominal concentration falls from $\sim 5\%$ to $\sim 1.7\%$ per atom at $n = 4$. Tetramer α , gaps, and Supplemental Material IPR/ J trends are therefore not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ converged neutral tasks support the quantitative claims below (Table S2). Charged-state geometry optimizations (12/12 converged) and vertical single-points for Marcus reorganization energies follow the same functional and basis choices (PBE+D3, 400 Ry on the tetramer grid) and are reported separately in Supplemental Material Fig. S5 (eight vertical ENERGY tasks at the neutral geometry; 6/8 converged at revision; verified λ values SI-only); they are orthogonal to and do not enter the $\mathcal{S}$ definition in Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is the total-energy difference per dopant site at fixed strain *without* elemental chemical-potential corrections; it is *not* a thermodynamic standard formation enthalpy ($\sim$ eV-scale tabulated values) and is used only for relative ranking within this protocol. In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the

substitution-energy scale: at $n = 4$ and +3% strain, P has $|\mathcal{S}| = 23.7$ meV/atom while the per-atom substitution scale is $|E_{\text{sub}}| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_{\text{sub}}| \approx 16$ eV per P site, $n_{\text{dop}}=12$, Table S1)—i.e. $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}} \sim 3\%$, large enough to shift *relative* stability margins among chemistries with comparable $|E_{\text{sub}}|$ in sequential doping-then-strain screens, even though $|\mathcal{S}|$ exceeds the rigid per-atom strain increment ($\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$ meV/atom for P) and does not reorder widely separated E_{sub} ranks (e.g., B vs. P at $\epsilon=0$).

Non-additive coupling metric. We quantify strain-doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

To first order in strain, $\mathcal{S}(\epsilon, \delta) \approx (\alpha_{\delta} - \alpha_0)\epsilon$ when $\alpha_{\delta} = dE/d\epsilon$ is evaluated for the doped cell and α_0 for the pristine reference at the same n (rigid-strain protocol); this links the mechanical coefficient α to the cross term at small ϵ .

Periodic-cell $\mathcal{S}$ values are extracted from converged CP2K total energies by in-house post-processing (Supplemental Material Table S2); the fifteen-point +3% strain audit is provided with the open data release. We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Supplemental Material Fig. S6). Following Capobianco *et al.* [4], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR is $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75-81$ meV reported for pristine qHP networks [4], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding (Supplemental Material). Marcus reorganization energies follow the vertical–adiabatic decomposition [4]: $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$, with separate internal ($\lambda_{\text{loc}}^{\pm}$) and environmental ($\lambda_{\text{env}}^{\pm}$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5. Twelve charged geometry optimizations supply adiabatic IP/EA; vertical single-points at

the neutral geometry are completing $\lambda^\pm$ (6/8 converged at revision) (Supplemental Material Fig. S5).

E. Validation benchmarks (Supplemental Material)

The full calculation map (converged tasks, main-text usage, and pending controls) is in Supplemental Table S2; ionic-relaxation energies (Table S3) and the alternate seed 137 grid (Table S4) appear there when complete. Three supplemental benchmarks—rigid vs. relaxed tetramer $\mathcal{S}_P$ (Table S3), alternate dopant seed 137 (Table S4), and a matched 400 Ry $n=6$ N control (Table S2)—gate interpretation of the N $\mathcal{S}(n)$ sign change cited in Secs. III C and IV. Marcus charged-state diagnostics (Fig. S5; 12/12 GEO_OPT converged; vertical λ in progress, 6/8 at revision) are orthogonal to the $\mathcal{S}$ audit and reported only in Supplemental Material. Substitution energy E_{sub} is *not* a thermodynamic formation enthalpy: it omits elemental chemical potentials and reports $(E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at fixed strain only.

III. RESULTS

We organize the results around total-energy non-additivity: Sec. III A summarizes band-edge context; Sec. III B reports linear strain coefficients and E_{sub} ; Sec. III C gives $\mathcal{S}(n)$ at +3% strain across dopants and supercell sizes. Marcus schematics and IPR/ J diagnostics are orthogonal to this audit and appear only in Supplemental Material Fig. S5–S6; adiabatic IP/EA and any verified λ values are in Supplemental Fig. S5 only—none are quoted in Sec. III—from twelve converged charged geometry optimizations with vertical single-points in progress (6/8 at revision).

A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). B compresses the π gap toward zero at $\epsilon=0$; N and P retain finite gaps; tensile strain narrows the pristine gap. These band-edge data contextualize total-energy $\mathcal{S}$ but are not used to infer $\mathcal{S}$.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens an order-of-magnitude spread in α : N (−303.2 meV/%, tension stabilizes the doped cell), B (+56.9 meV/%, tension destabilizes), and P (−19.1 meV/%, weakly stabilizing) (Table S1). The opposite sign for N vs. B persists across the $\epsilon \in \{-5, \dots, +5\}$ % grid and is not an artifact of a single strain point. At $\epsilon = +5\%$, N remains the most

tension-stabilized dopant (α most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative α , foreshadowing why its linear coefficient poorly predicts $|\mathcal{S}| = 23.7$ meV/atom for P at $n=4$ under +3% strain (Sec. III C). Substitution energies at $\epsilon = 0$ span $E_{\text{sub}} = +73.4$ eV/dopant (B), -115.3 eV/dopant (N), and -15.8 eV/dopant (P)—raw total-energy differences without chemical-potential reference, not standard formation enthalpies—with $n_{\text{dop}} = 12$ –13 counted from input coordinates (Table S1). Figure 1(c) highlights the α split; ranking dopants by $\epsilon = 0$ $|E_{\text{sub}}|$ alone is misleading once +3% strain is applied.

C. Non-additive strain–doping coupling

Figure 1(d) tracks $\mathcal{S}(n)$ with one dopant per C_{60} at +3% biaxial strain. At $n = 4$ (seed 42 periodic placement; Table S4 pending), $\mathcal{S} = -6.2$ meV/atom (B), $+3.0$ meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the tetramer grid (Table S1). Across fifteen coupled points ($n \in \{1, 2, 4, 6, 8\}$, B/N/P), $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom at larger n for B and P, whereas N is smaller in magnitude at $n = 4$ but comparable at $n = 1$. For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (−2.3 meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until the matched 400 Ry $n=6$ control in Supplemental Table S2 and Tables S3–S4 complete (Sec. II E). Core size-scaling claims focus on $n \leq 4$ at matched 400 Ry cutoffs; B and P show monotonic $|\mathcal{S}|$ decrease from $n = 1$ toward $n = 4$, whereas $n \geq 6$ trends are reported for completeness only and mix the 350 Ry production cutoff. We report provisional $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$ fits in Supplemental Material but withhold asymptotic $\mathcal{S}_\infty$ claims until cutoffs are matched across n . All periodic $\mathcal{S}$ values quoted in this subsection are tabulated in Supplemental Material (fifteen $n \times$ dopant points at +3% strain). Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

IV. DISCUSSION

Context. Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–14, 19], the distinction here is a same-functional periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] on a rigid-strain upper-bound protocol. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [19]; isomer-engineered C_{70} network designs similarly tune band edges through topology choice rather than joint substitutional–

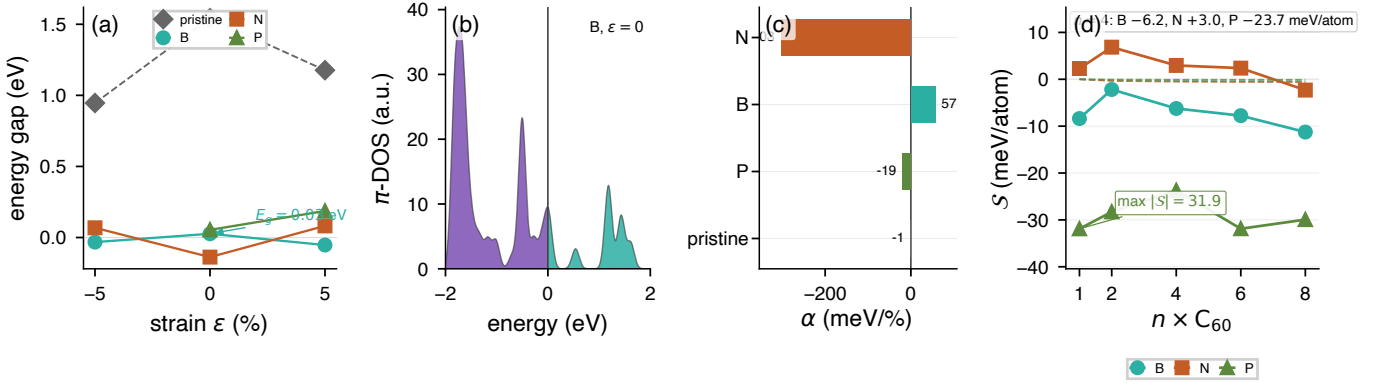


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO-LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (color scheme follows Ref. 4). (c) Mechanical α (Table S1). (d) $S(n)$ at +3% (max $|S|=31.9$ meV/atom for P @ $n=1$; at $n=4$: -6.2, +3.0, -23.7 meV/atom for B/N/P); Marcus λ (partial, 6/8 at revision; Supplemental Fig. S5); IPR/ J in Supplemental Material Fig. S6.

mechanical load [20]; we instead quantify the omitted additive cross term S in total energy. Band-edge, spin-resolved, and hopping metrics (Supplemental Material Fig. S6) contextualize mechanism but are not used to infer S .

Mechanistic synthesis (three coupling classes).

(i) **Opposite α for N vs. B.** N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α)—consistent with donor vs. acceptor π -counting, the gap-vs-strain slopes in Fig. 1(a), and dopant-resolved π -DOS in Fig. 1(b) and Supplemental Fig. S4. (ii) **Large $|S|$ for P despite small $|\alpha|$.** P carries the largest covalent-radius excess (+30 pm vs. C; Table S5) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta\bar{d} \approx 0.00003$ Å from $\epsilon=0$ to +3% vs. +0.0085 Å for B; Table S5). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both knobs are engaged—consistent with P’s intermediate E_{sub} yet the largest $|S|$ at $n=4$ (23.7 meV/atom at $n=4$; Supplemental Table S2) despite weak tetramer $|\alpha|$. This closes the strain-response foreshadow in Sec. IIIB: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term. (iii) **N S sign change with n .** With one dopant per C_{60} , electrostatic fields dilute at large n ; any N sign flip between $n=6$ and $n=8$ is confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. IIE). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n).

(iv) **Beyond first-order $(\alpha_\delta - \alpha_0)\epsilon$.** A one-parameter mechanical estimate $S \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ uses tetramer α at 5% doping (Table S1) and is *not* matched to the pe-

riodic $n=4$ concentration ($\sim 1.7\%$ per atom). We therefore use it only as a qualitative sanity check (concentration and boundary conditions differ) and not as proof of nonlinearity: at $\epsilon=+3\%$ it gives $\sim +0.7$, -3.8 , and -0.2 meV/atom for B/N/P, whereas the periodic $n=4$ rigid values are -6.2 , $+3.0$, and -23.7 meV/atom. The mismatch signals that the cross term is not reducible to linear strain sensitivity at a different concentration; P in particular requires explicit coupled evaluation.

(v) **Significance versus DFT uncertainty.** S is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. At $n=4$ and +3% strain, $|S| = 23.7$ meV/atom for P is $\sim 3\%$ of the per-atom E_{sub} scale (Table S1) but still $\gg$ rigid per-atom strain increments from α . Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

Design implications. At $n=4$ and +3% strain the coupled correction is $|S| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates misestimate the load-dependent *margin* by an order of magnitude even though widely separated E_{sub} ranks at $\epsilon=0$ (e.g., B vs. P; Table S1) are unchanged. Dopants with larger covalent-radius excess (P: +30 pm vs. B: +7 pm; Table S5) show the largest $|S|$, so *size-mismatched* substituents demand coupled (ϵ, δ) screening rather than one-parameter pipelines. The fifteen-point periodic grid tabulates $|S|$ at each (n, δ) for screening workflows under the rigid protocol (Supplemental Table S2). At +3% strain, P carries the largest additive-approximation error in S ($|S| = 23.7$ – 31.9 meV/atom from $n=4$ to $n=1$); co-design should treat P under tension as a *coupled* stability problem, not $|E_{\text{sub}}|$ at $\epsilon=0$ alone. Where $S < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $S > 0$ (N at $n=4$), sequential screens overestimate stability under load. For chemistries with comparable $|E_{\text{sub}}|$ or in multi-dopant

competition, omitting $\mathcal{S}$ can shift margins by several percent of the per-atom substitution scale—much larger than the rigid strain increment from α —even when the global dopant rank at $\epsilon=0$ is unchanged. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. For B at $n = 4$, both negative $\mathcal{S}$ and positive α imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating $\mathcal{S}$. $N \mathcal{S}$ vs. n is reported for completeness but excluded from design rules until Table S2–S4 controls converge.

Polaron transport (Supplemental Material only). The Marcus workflow in Fig. S5–S6 parallels Ref. 4—adiabatic charged-state energies plus vertical evaluations at the neutral geometry—at the uniform PBE+D3 level used for $\mathcal{S}$. Where that study linked hopping J and frontier-orbital localization to mobilities in pristine qHP networks, we retain the same observables for comparability (Supplemental Fig. S6) but treat reorganization energies as *orthogonal* to the neutral total-energy audit: adiabatic IP/EA and any verified λ appear only in Supplemental Fig. S5 (12/12 charged geometry optimizations converged; vertical single-points 6/8 at revision; none quoted in Sec. III) and do not feed back into $\mathcal{S}(n)$ in Fig. 1(d). Six of eight vertical tasks at revision populate hole and electron reorganization channels for separate dopant classes in Supplemental Fig. S5, validating the Capobianco-style workflow at the PBE+D3 level without using λ to rank neutral total-energy margins. Endohedral band-gap tuning [14] and substitutional $\mathcal{S}$ address a second orthogonal knob (cage filling vs. cage substitution) and are not interchangeable for sequential-screen corrections.

Limitations and outlook. Three protocol limits are actively benchmarked (Supplemental Table S2; Sec. II E): rigid strain (Table S3), fixed-seed ensembles (Table S4), and mixed 400/350 Ry cutoffs across n (Table S2 cutoff400 control). Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10; $\mathcal{S}_{\text{gap}}$ not quantified; Mayer bond-order and Bader charge partitioning along the strain path are not computed (Table S5 reports geometric metrics only); Marcus λ awaits completion of eight vertical single-points (6/8 converged at revision; 12/12 charged geometry optimizations converged); mobility tensors are not computed—polaron transport anisotropy in related fullerene networks is treated only comparatively [4, 6] (Supplemental Material Fig. S5–S6).

V. CONCLUSIONS

Sequential strain-doping workflows in covalent molecular networks can omit a meV/atom-scale total-energy

cross term when local dopant distortion couples non-linearly to external biaxial load. In qHP C_{60} graphullerene we quantified this omission with $\mathcal{S}$ under a uniform PBE+D3 rigid-strain protocol, addressing the three Introduction questions: donor N and acceptor B show opposite α ; joint (ϵ, δ) load produces $|\mathcal{S}|$ at the $\mathcal{O}(10)$ meV/atom level on a fifteen-point periodic grid at +3% strain (seed 42 placement); and the largest $|\mathcal{S}|$ does not track the smallest tetramer $|\alpha|$ —size-mismatched P couples through pre-existing cage distortion (Table S5) rather than linear elasticity alone. Reported $\mathcal{S}$ values are *upper bounds* under fixed fractional coordinates; ionic-relaxation, alternate-seed, and matched-cutoff controls (Tables S2–S4) will test whether the cross term remains design-relevant after relaxation and whether rankings are seed-robust. Additive estimates can mis-estimate *relative* stability margins among competing chemistries at the few-percent level of the substitution-energy scale without reordering widely separated dopant ranks. Marcus λ and hopping metrics (Supplemental Material Fig. S5–S6) are orthogonal to the $\mathcal{S}$ audit and do not alter the neutral-energy conclusions in Fig. 1(d); vertical single-points were in progress (6/8 converged at revision), with verified λ reported only in Supplemental Material and partial completion already exercising distinct hole and electron channels without changing $\mathcal{S}(n)$ rankings. Coupled (ϵ, δ) screening is therefore advisable for size-mismatched substituents and for close substitution-energy competitors, complementing strain-only mechanical [12], derivative-network [11], and endohedral band-edge [14] tuning axes. Future work will extend the seed 137 placement audit to periodic cells, add Mayer bond-order and Bader partitioning along the strain path, and complete the eight vertical Marcus tasks in Supplemental Fig. S5.

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DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering> (repository *sci-simukit*), including converged CP2K inputs and outputs, machine-readable synergy and polaron audit tables, and scripts to regenerate the main and Supplemental Material figures. Pending ionic-relaxation, alternate-seed, and matched-cutoff validation tasks are documented in Supplemental Tables S2–S4. A tagged release matching the manuscript revision will be issued upon acceptance.

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